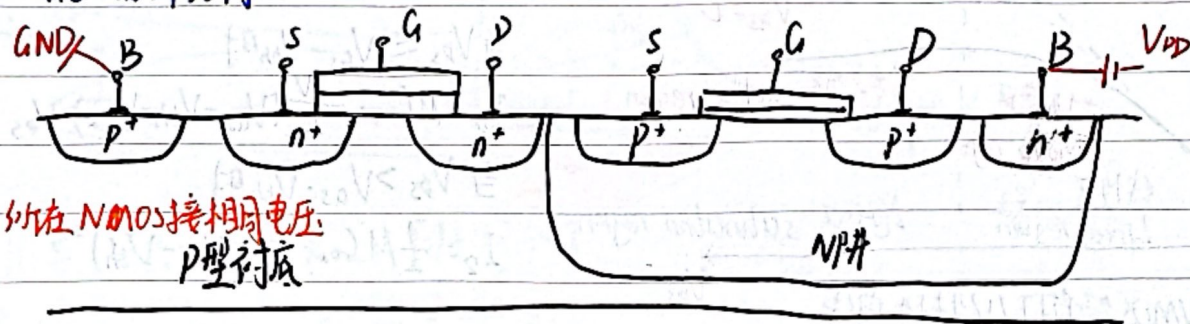


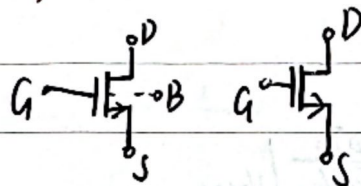
CMOS器件原理

1. CMOS器件结构

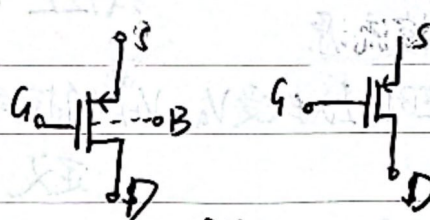


四个管脚：栅极(G)，漏极(D)，源极(S)，衬底(Bulk, B)

图示：



NMOS



PMOS

工作原理(以NMOS为例)

当在NMOS栅极上施加一定大小的正偏电压时，会在栅极下方衬底处形成电场吸引电子，当 $V_g < V_{th}$ ($V_{threshold}$ ，简称 V_{th} 或 V_b ，阈值电压)时，被吸引电子中和部分空穴，形成一个中和区，称为耗尽区；当 $V_g > V_{th}$ 时，多余自由电子形成一条导电沟道，称为反型层。

PMOS类似，极性相反

2. MOS器件的I/V特性

当 $V_{ds} > 0$ 时，从漏极到源极出现电势降，反型层各个部分吸引电子的能力从源极到漏极逐渐下降，从而形成厚度不一致的导电沟道

I/V特性

根据电流计算式： $I_D = WQ_n v$ $v = ME$ 载流子迁移速度 W 为栅宽

设漏源方向长度为 x 的位置电势为 $V_{ch}(x)$ ，则 $E = \frac{dV_{ch}(x)}{dx}$

$$\therefore Q_n = C \cdot \phi_V$$

$$Q_n = C_{ox} (V_{gs} - V_{ch}(x) - V_{th})$$

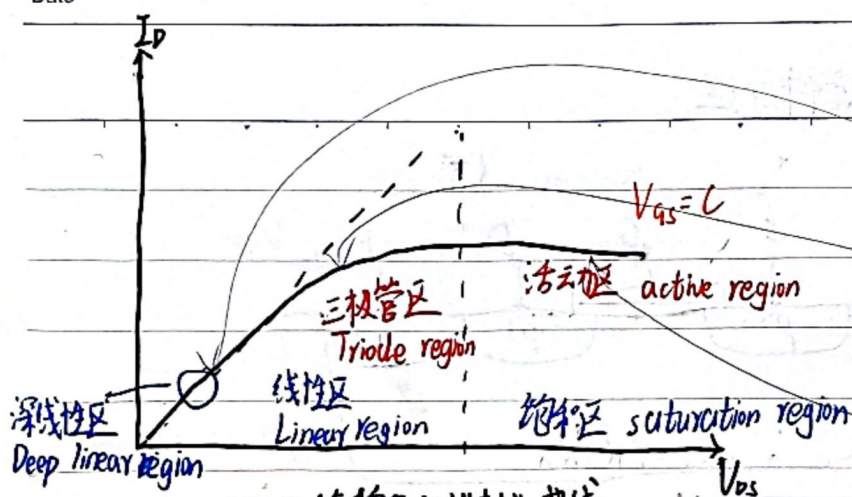
C_{ox} 为单位面积电容

积分得 $I_D = \mu C_{ox} \frac{W}{L} \cdot (V_{gs} - V_{th} - \frac{V_{ds}}{2}) V_{ds}$

当 $V_{ds} > V_{gs} - V_{th}$ 时，不应减小而趋于饱和

因此：





NMOS管的I/V特性曲线

当 $V_{DS} < V_{GS} - V_{th}$ 时

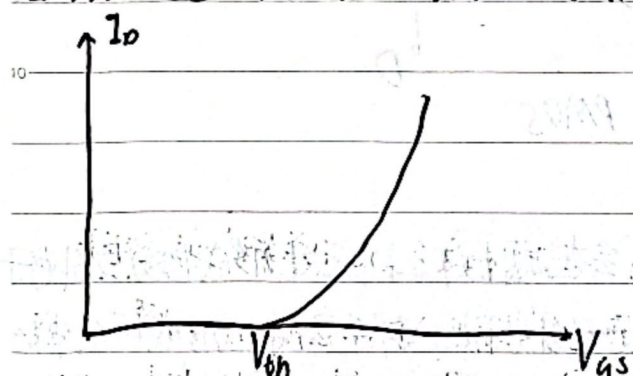
$$I_D = \mu C_{ox} \cdot \frac{W}{L} (V_{GS} - V_{th}) \cdot V_{DS}$$

当 $V_{DS} \leq V_{GS} - V_{th}$ 时

$$I_D = \mu C_{ox} \cdot \frac{W}{L} (V_{GS} - V_{th} - \frac{V_{DS}}{2}) \cdot V_{DS}$$

当 $V_{DS} > V_{GS} - V_{th}$ 时

$$I_D = \frac{1}{2} \mu C_{ox} \cdot \frac{W}{L} (V_{GS} - V_{th})^2$$

饱和区 $\approx$ VCCS: 压控流源线性区 $\approx$ 当 $V_{DS} = C$ 且工作在饱和区时, I_D 仅受 $V_{GS} - V_{th}$ 的控制, 即定义跨导 $g_m = \frac{\partial I_D}{\partial V_{GS}} \bigg|_{V_{DS}=C}$

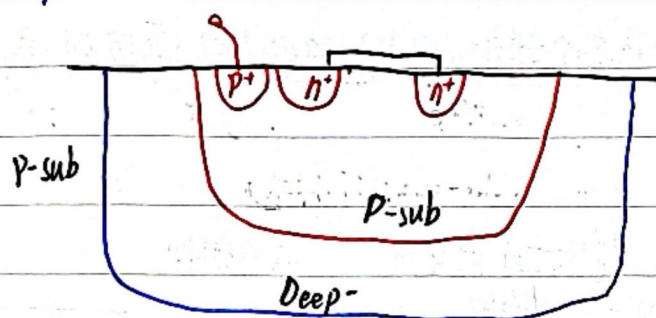
$$= \mu C_{ox} \cdot \frac{W}{L} (V_{GS} - V_{th}) = \frac{2I_D}{V_{GS} - V_{th}} = \sqrt{2\mu C_{ox} \frac{W}{L} I_D}$$

$$\text{线性区 } g_m = \frac{1}{R_{on}} = \mu C_{ox} \cdot \frac{W}{L} (V_{GS} - V_{th} - \frac{V_{DS}}{2})$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \bigg|_{V_{DS}=S}$$

PMOS类似, 极性相反

3. Deep-Well



NMOS可接不同电压

4. CMOS器件的小信号模型

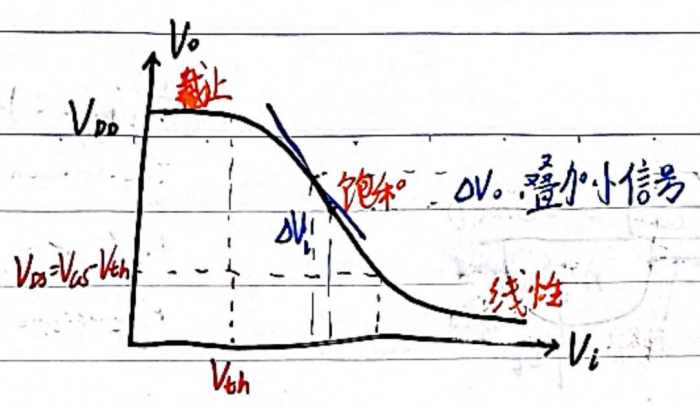
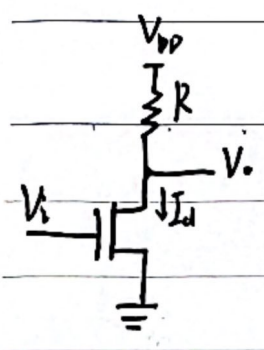
Large Signal

$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_i - V_t)^2$$

$$V_o = V_{DD} - \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_i - V_t)^2 \cdot K$$

饱和区





可推导 $\Delta V_O = -\frac{2I_D}{V_{OV}} R \cdot \Delta V_i \left[1 + \frac{\Delta V_i}{2V_{OV}} \right]$ 忽略

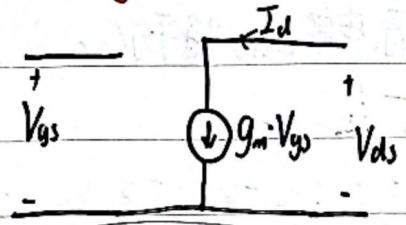
$$V_{OV} = V_{GS} - V_{th}$$

当 $\Delta V_i \ll 2V_{OV}$ 时 $\Delta V_O \approx -\frac{2I_D}{V_{OV}} R \cdot \Delta V_i$

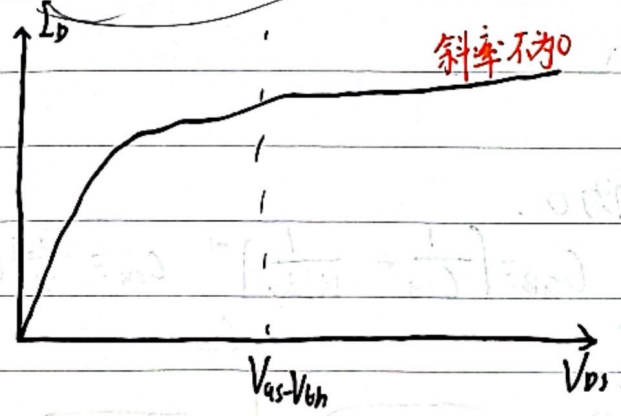
$$\therefore \frac{dV_O}{dV_i} \bigg|_{V_i=V_{GS}} = -\frac{2I_D}{V_{OV}} R \quad \text{即} \quad \frac{V_O}{V_i} = -g_m R$$

$$g_m = \frac{2I_D}{V_{OV}} = \mu \cdot C_{ox} \cdot \frac{W}{L} (V_{GS} - V_{th})$$

Small-signal Model



沟道调制效应



当NMOS管工作于饱和区时, I_D 实际与 V_{DS} 弱正相关
这一结果是包括漏端引入的势垒降低、衬底电流、诱导体效应等多种物理效应综合作用的结果, 我们将这一系列效应的综合作用称作“沟道长度调制效应”

假定 I_D 与 V_{DS} 线性正相关, 引入沟道长度调制系数 λ , $\lambda \propto \frac{1}{L}$ 沟道长度, 则 g_m 与 I_D 可被修正为

$$g_m = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th}) (1 + \lambda V_{DS})$$

$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS})$$

则此时 $g_{ds} = \frac{dI_D}{dV_{DS}} = \frac{\lambda I_D}{1 + \lambda V_{DS}} \approx \lambda I_D$ 漏源电导

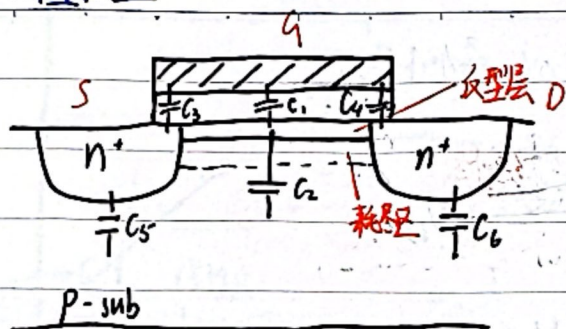
$$\frac{1}{\lambda I_D} \propto \frac{1}{L}$$

在信号模型中可在漏源两端接一个阻值为 $\frac{1}{\lambda I_D}$ 的电阻 r_o , 称为本征阻抗

增大 L , 会减小沟道调制效应



6. 本征电阻模型



工作于线性区MOS管的器件电容

记 C_{ac} 为栅极与近似均匀的导电沟道构成的平行板电容器, 则

$$C_{ac} = WL \frac{\epsilon_{ox}}{t_{ox}} = WL C_{ox}$$

此时, 由于 V_{as} 较小, 近似认为 $V_D \approx V_S$, 则栅漏电容 C_{gd} 与栅源电容 C_{gs} 将平分 C_{ac}

$$C_{gd} = C_{gs} = \frac{1}{2} C_{ac} = \frac{1}{2} WL C_{ox}$$

此时忽略沟道与衬底电容 C_{cb}

工作于饱和

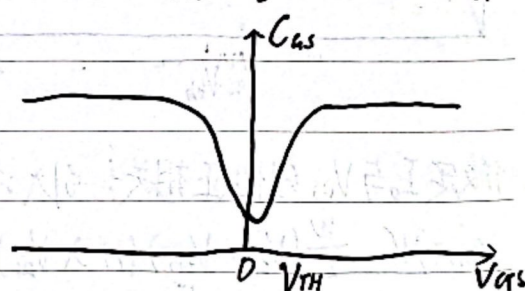
此时导电沟道被夹断 $C_{gd} \approx C_{gb} \approx 0$ $C_{gs} = \frac{2}{3} WL C_{ox}$

截止

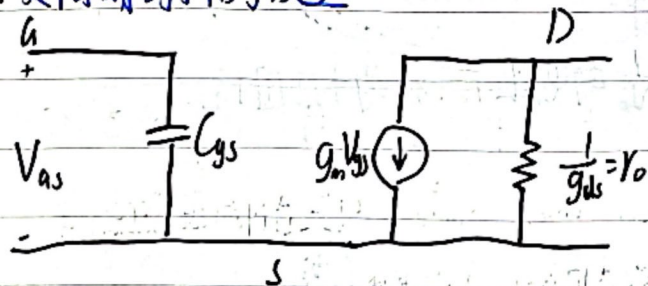
此时无导电沟道, 栅漏电容 C_{gd} 与栅源电容 C_{gs} 均为 0.

由于无反型层屏蔽作用, 栅极与衬底间电容不再为 0 $C_{gb} = \left[\frac{1}{C_{cb}} + \frac{1}{WL C_{ox}} \right]^{-1}$ $C_{cb} = \frac{C_{gs}}{x_d} WL$

	截止	线性	饱和
C_{gs}	0	$\frac{1}{2} WL C_{ox}$	$\frac{2}{3} WL C_{ox}$
C_{gd}	0	$\frac{1}{2} WL C_{ox}$	0
C_{gb}	$\left[\frac{1}{C_{cb}} + \frac{1}{WL C_{ox}} \right]^{-1}$	0	0



7. 更精确的小信号模型

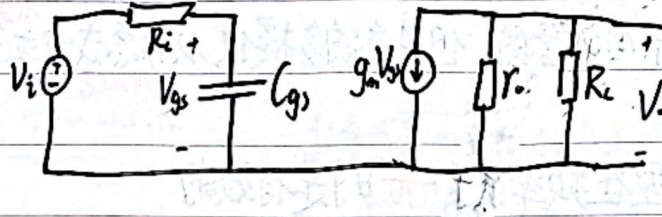
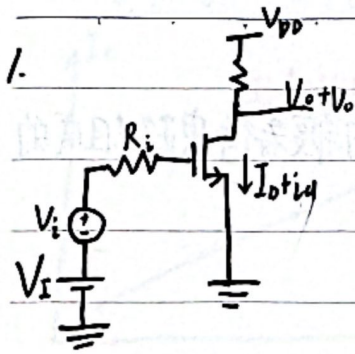


$$g_m = \frac{2I_D}{V_{ov}}$$

$$C_{gs} = \frac{2}{3} WL C_{ox}$$

$$r_o = \frac{1}{\lambda I_D}$$

共源集成放大器



$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{g_m (Y_o // R_L)}{1 + s R_i C_{GS}} \quad \text{DC 直流}$$

当 $R_L \ll r_o$ 时 $A_{DC} = g_m R_L$ 直流增益

$$\therefore I_D = \frac{1}{2} g_m V_{OV} \quad g_m = \frac{A_{DC}}{R_L}$$

$$V_{DD} I_D = V_{DD} \cdot \frac{1}{2} g_m V_{OV} = V_{DD} \cdot \frac{1}{2} \frac{A_{DC}}{R_L} V_{OV}$$

$$\therefore P = V_{DD} I_D = \frac{1}{2} \frac{V_{DD}}{R_L} A_{DC} V_{OV} \quad \frac{1}{R_i C_{GS}} = \frac{1}{R_i \frac{2}{3} \mu C_{ox} L}$$

带宽 $W_{3dB} = \frac{1}{R_i C_{GS}} \quad C_{GS} = \frac{2}{3} W L C_{ox}$

$$\downarrow g_m = \mu C_{ox} \frac{W}{L} V_{OV} \quad g_m = \frac{A_{DC}}{R_L}$$

$$W_{3dB} = \frac{3}{2} \frac{R_L}{R_i} \cdot \frac{1}{A_{DC}} \cdot \frac{\mu}{L^2} V_{OV}$$

$$= \frac{3}{2} \frac{\mu V_{OV}}{R_i g_m L^2}$$

$$= \frac{3}{2} \frac{R_L}{R_i} \frac{\mu V_{OV}}{A_{DC}} \cdot \frac{1}{L^2}$$

2. CMOS器件的特性参数分析

设计目标:

Lots of g_m without investing much current.

低功耗

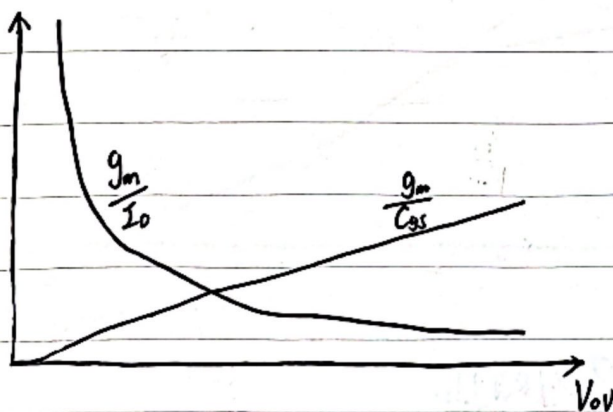
Lots of g_m without having large C_{GS}

宽带宽

引入

$$\frac{g_m}{I_D} = \frac{2}{V_{OV}}$$

$$\frac{g_m}{C_{GS}} = \frac{3}{2} \frac{\mu V_{OV}}{L^2}$$



可得出, 随 V_{OV} 上升, $\frac{g_m}{I_D}$ 与 $\frac{g_m}{C_{GS}}$ 的大小并不呈相同的变化趋势, 因此 tradeoff 权衡取舍

而

$$\frac{g_m}{I_D} \cdot \frac{g_m}{C_{GS}} = \frac{3\mu}{L^2}$$

因此 L 越小有助于使电路朝更低功耗与更宽带宽的高性能发展



传输频率 ω_T : MOS管的电流增益 $\frac{I_o}{I_i}$ 下降到1的角频率

即 $\frac{V_{gs}}{I_i} = g_m V_{gs}$ $\therefore \omega_T = \frac{g_m}{C_{gs}}$ $f_T = \frac{\omega_T}{2\pi}$

ω_T 的值并不能准确表示 MOS管的, 但其能够提供从集总或分立元件的角度看待电路组成的频率分界参考。
不能单纯用 RLC 表征
能用 RLC 表征

一般而言, 集总参数模型在频率低于 $\frac{\omega_T}{10}$ 时是有效的

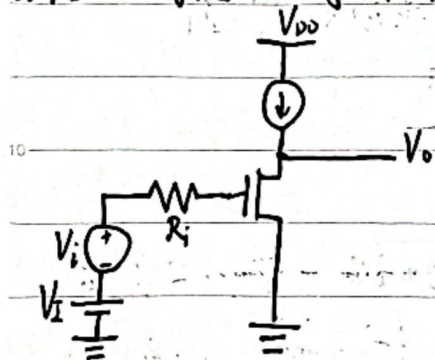
本征增益 $g_m r_o$: 描述 MOS管放大能力的参数

已知直流增益 $A_{DC} = g_m (R_L \parallel r_o)$, 当 $R_L \ll r_o$ 时, $A_{DC} \approx g_m R_L = \frac{2I_o R_L}{V_{ov}}$

此时为保证工作在饱和区, 至少需要满足 $V_o > V_{ov}$, 即 $I_o R_L < V_{ov}$ $\therefore A_{DC} < \frac{2V_{ov}}{V_{ov}} = 2$

当 $R_L = r_o$ 或 $R_L > r_o$ 时, r_o 作用开始显现, 电路放大能力被 r_o 限制, 此时用一电流源代替 R_L

当 R_L 趋于无穷大的过程中, 电路的电压增益逐渐收敛于 $g_m r_o$ 这一值, 称为 **本征增益**

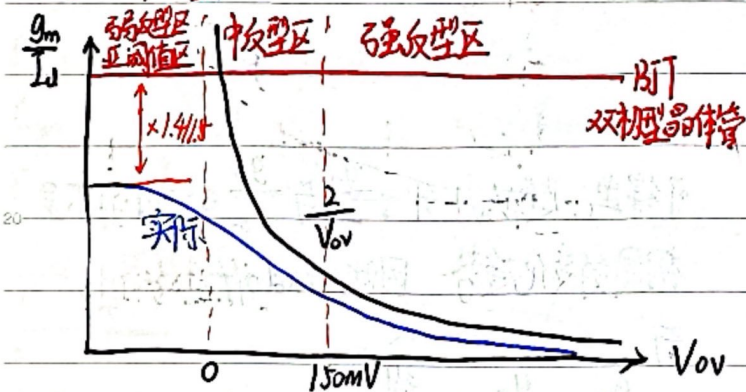


对于高阻点, 直流工作点不太稳定

假设所研究的 MOS管为长沟道模型, 则可知该三个参数与 W 无关, 减少了电路设计时需考虑的变量

3. CMOS的工作区域

跨导效率 $\frac{g_m}{I_D}$



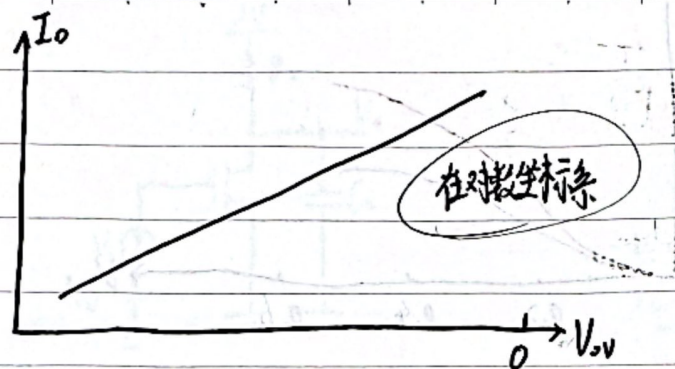
弱反型区 (亚阈值区)

实际器件工作于 $V_{ov} < 0$ 时, 实际上仍存在微小但不为0的 I_D

重新定义 V_{th} : 恰好使少数载流子能够中和衬底多子对应的 V_{gs} 值



此时电流为扩散电流



此时可知, 电流与 V_{GV} 成指数关系, 可等效为 BJT

此时 $I_D \approx I_0 \cdot e^{\frac{V_{GV}}{\eta V_T}}$

其中 η 是一个与 MOS 工艺相关的常数

$\eta = \frac{C_{js} + C_{ox}}{C_{ox}} = 1 + \frac{C_{js}}{C_{ox}} \approx 1.4$

中反型区

这一区域中, 漂移电荷与扩散电荷同时承担导电沟道的任务。

漂移电荷: $v = \mu E$, 扩散电荷: $v = \frac{kT}{q} \mu \frac{dn}{dx}$, 则扩散电流与漂移电流比值:

$\frac{\text{漂移电流}}{\text{扩散电流}} \propto \frac{V_{GS} - V_{th}}{\frac{kT}{q}}$ 当 V_{GV} 大于 $\frac{kT}{q}$ 时, 漂移电流占主导地位

强反型区 $V_{GV} > 150mV$

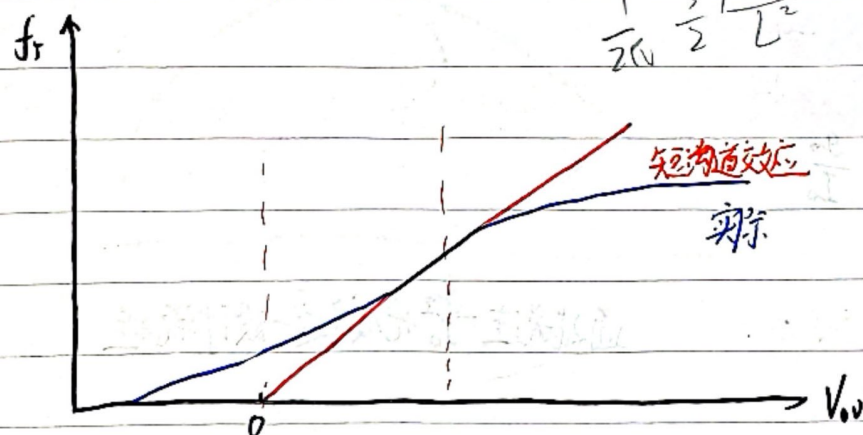
长沟道模型

由图可知, 在强反型区, 实际曲线与平方律特性曲线基本相同, 数值上则是平方律的 80%~90%。这是由于强反型区导电沟道处建立的高水平方向电场引起的载流子漂移速度饱和与高垂直方向电场引起的载流子漂移能力下降等多种物理效应。

短沟道效应

结论: 使用平方律长沟道器件模型并不能精确描述 MOS 器件 $\frac{g_m}{I_D}$ 特性, 且误差随 L 减小变得更大。因此在设计时直接选用 $\frac{g_m}{I_D}$ 作为新设计变量

4. W_T (或 f_T) 与 V_{GV} 之间的关系



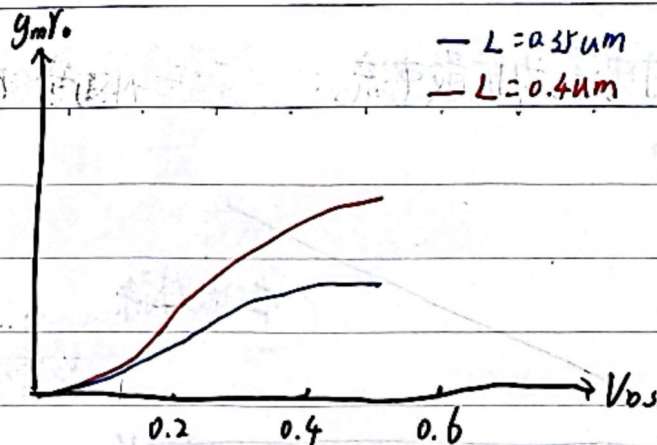
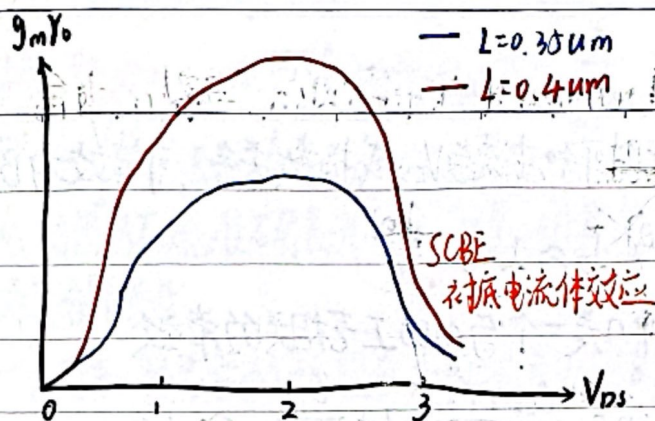
长沟道: $f_T = \frac{1}{2\pi} \frac{3}{2} \frac{\mu V_{GV}}{L^2}$

在正阈值区, MOS 管具有双极型晶体管行为

在强反型区, 受到短沟道效应

5. g_m/I_D 与 V_{GS} 之间的关系

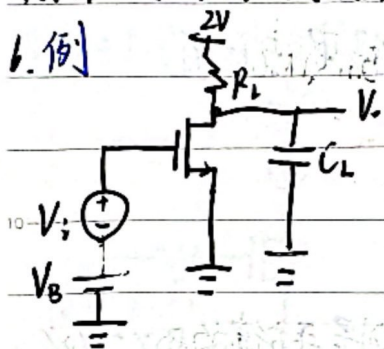




由于 r_o 的变化而显著变化

V_{ov} 并不是一个有效的设计参数, V_{ov} 也不等于 V_{dset} 漏极饱和电压

6. 例



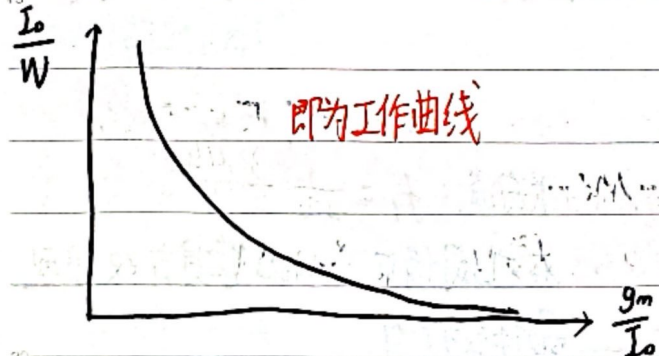
给定 $A_{dc} = -2$, $I_o \leq 2mA$, $f_{-3dB} = 100MHz$, $C_L = 1pF$.
要求 MOS 管尺寸尽可能小

$\therefore g_m r_o > 0$ $\therefore A_{dc}$ 主要受 R_L 影响, 则根据 $f_{-3dB} = \frac{1}{2\pi R_L C_L}$ 可求 R_L

$A_{dc} = -g_m R_L$ 求 g_m 则可求 $\frac{g_m}{I_o}$

$\therefore g_m = \mu C_{ox} \frac{W}{L} V_{ov}$ $\therefore W \sim \frac{1}{V_{ov}} \sim \frac{g_m}{I_o}$, 因此 I_o 越大, W 越小, 又可求 I_o

此时可用长沟道模型, 但并不准确, 因此通过 $\frac{g_m}{I_o}$ 求 W



由 $\frac{g_m}{I_o}$ 找对应点, 直接 $I_o / \frac{I_o}{W}$ 求 W

7. 设计流程

Design

specifications
 $\downarrow$
 $g_m, \frac{g_m}{I_o}, f_t, \dots$

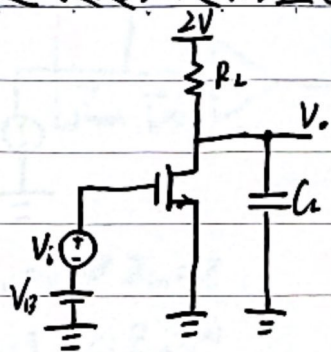
通过关注 $\frac{g_m}{I_o}$ 完成了这一设计流程

Implementation/
Verification

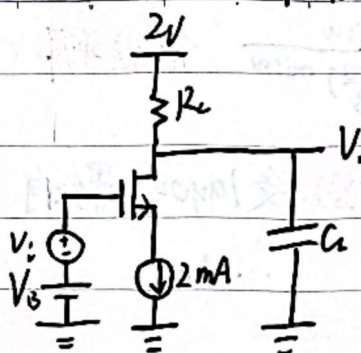
determine device size
 $\downarrow$
simulation results



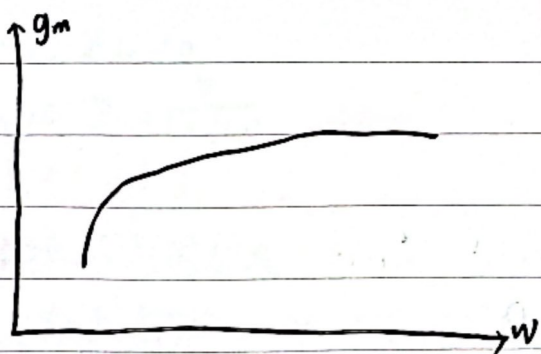
另外, 还可以通过给定偏置电流, 直接扫描器件宽度 W 的方法来确定管子尺寸, 直到 g_m 符合



$\Rightarrow$



$$2I_0 \mu C_{ox} \frac{W}{L}$$



由于 I_0 固定, $I_0 = \frac{1}{2} \mu C_{ox} \frac{W}{L} V_{ov}^2$

$\therefore W \sim V_{ov}^{-2}$

$\therefore g_m = \mu C_{ox} \frac{W}{L} V_{ov} \therefore g_m \sim \sqrt{W}$

$\therefore g_m$ 与 W 近似满足 1/2 次方关系

8. 改变沟道长度 L 的目的:

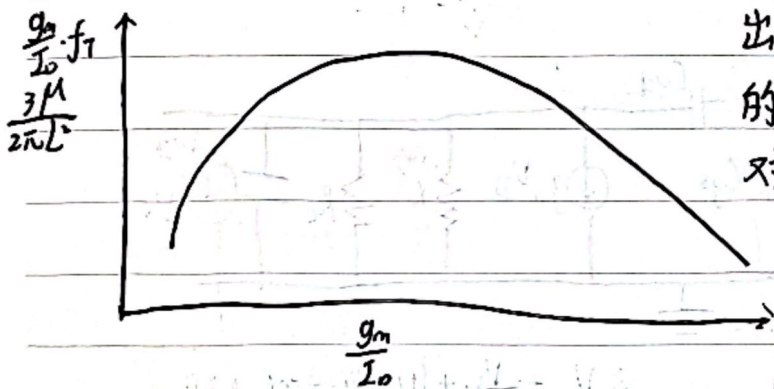
① 更好的匹配性

② 更大的增益. $\therefore \gamma_0 \sim L$ $\therefore$ 增大 L 可获得更高的本征增益

③ 更小的漏电流 在对亚阈值区电导的分析可知, 漏电流与沟道长度成反比

④ 应用于超低功耗 (nW) 设计.

9. 工艺热点:



出发点让 MOS 管偏置在“工艺热点”(g_m/I_0 的最大值)附近,

对 $L = 0.35 \mu m$ 工艺而言, V_{ov} 大约 200mV

10. 本征电容

Overlap capacitance: 栅极与漏源极重叠形成的电容

Two components

- direct overlap $\sim C_{ox} W L_{overlap}$ - 侧壁电容



Area (A_s and A_D) and Perimeter (P_s , P_D)

$$C_{jdb} = \frac{A_s \cdot C_j}{(1 + \frac{V_{DS}}{P_B})^{m_j}} + \frac{P_s \cdot C_{jsw}}{(1 + \frac{V_{DS}}{P_B})^{m_{jsw}}} \quad \text{很难计算}$$

junction capacitance (结电容): 受 layout 影响

$$C_j \approx 0.8 C_{gs}$$

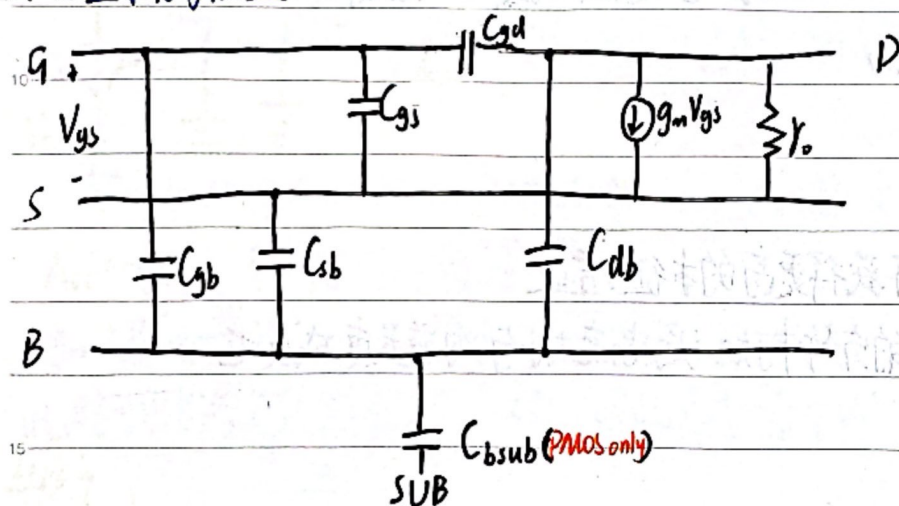
11. PMOS Well Capacitance

N阱在PMOS管中会和衬底形成一个PN结

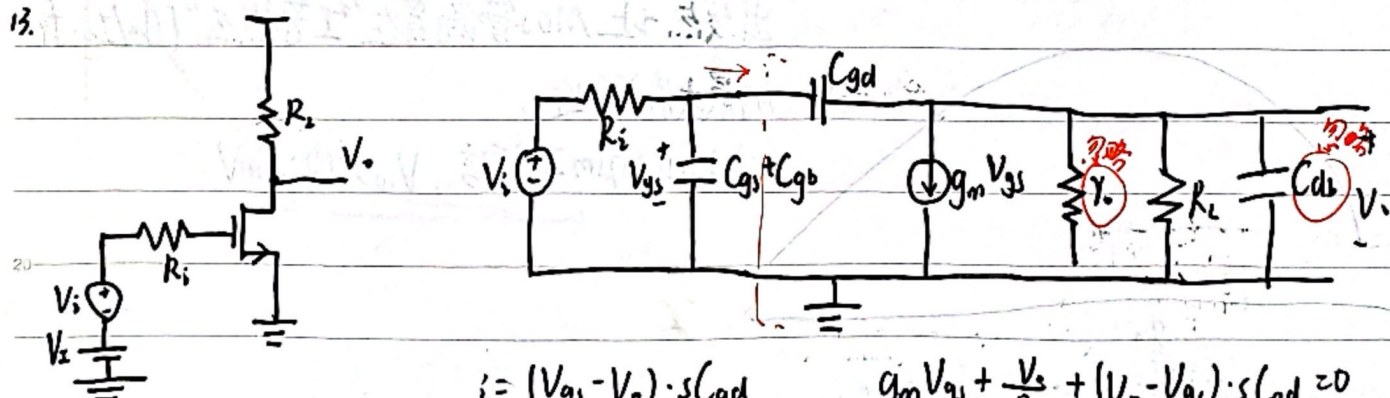
- 当 $N\text{-well} = V_{DD}$, $Sub = GND$ 时可看作AC短路

- 当 $N\text{-well}$ 与源极相连时与 GND 之间存在一个电容

12. 完整小信号模型



13.



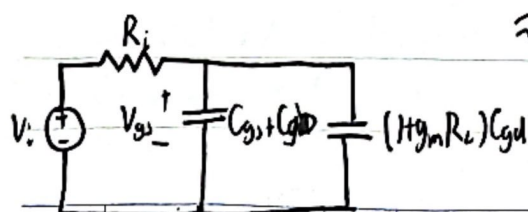
$$i = (V_{gs} - V_o) \cdot s C_{gd}$$

$$g_m V_{gs} + \frac{V_o}{R_L} + (V_o - V_{gs}) \cdot s C_{gd} = 0$$

$$\therefore Y(s) = \frac{i(s)}{V_{gs}(s)} = [1 - A_v(s)] \cdot s C_{gd}$$

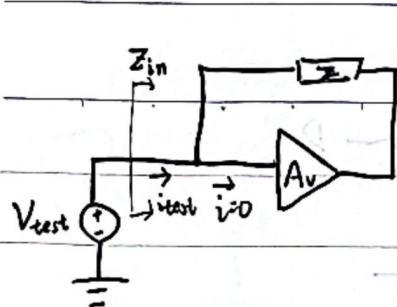
$$A_v(s) = -g_m R_L \left[\frac{1 - s \frac{C_{gd}}{g_m}}{1 + s R_L C_{gd}} \right]$$

$$\approx (1 + g_m R_L) \cdot s C_{gd} \quad \text{密勒效应}$$



$$f_{-3dB} \approx \frac{1}{2\pi R_L [C_{gs} + C_{gd} + (1 + g_m R_L) C_{gd}]}$$



$C_L \perp C_{in}$


$$Z_{in} = \frac{V_{test}}{i_{test}} = \frac{V_{test}}{\frac{V_{test} - A_v V_{test}}{Z}} = \frac{Z}{1 - A_v}$$

$$Y_{in} = Y(1 - A_v)$$

$$- A_v = 0 \Rightarrow Z_{in} = Z$$

$$- A_v = 1, \Rightarrow Z_{in} = \infty$$

$$- A_v > 1, Z_{in} < 0$$

$$- A_v < 0, Z_{in} = \frac{Z}{1 + |A_v|} \quad \text{阻抗减小}$$

当考虑负载电容时: 密勒逼近不可用

① 只能在低频时放大

② 3-dB 取代于 C_L

③ 带宽被 C_L 限制

4. Zero Value Time Constant (ZVTC) Analysis (零值时间常数分析) 用于分析主极点位置

① 只留一个电容, 短路所有独立压源, 移除所有独立流源

② 计算电容所连的阻抗, 得到 $\tau_j = R_j C_j$

③ 对电路中所有电容重复

④ 将所有 τ 相加, $\omega_{-3dB} \approx \frac{1}{\sum \tau_j}$

共栅放大电路

1. 体效应:

① 当 $V_{SB} > 0$ 时, 耗尽层变厚

② 耗尽层会捕获源极电子, 需要更大的 V_{GS} 才能补偿这种影响

会导致门限电压 V_t 增大, $V_t = V_{t0} + \gamma(\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f})$

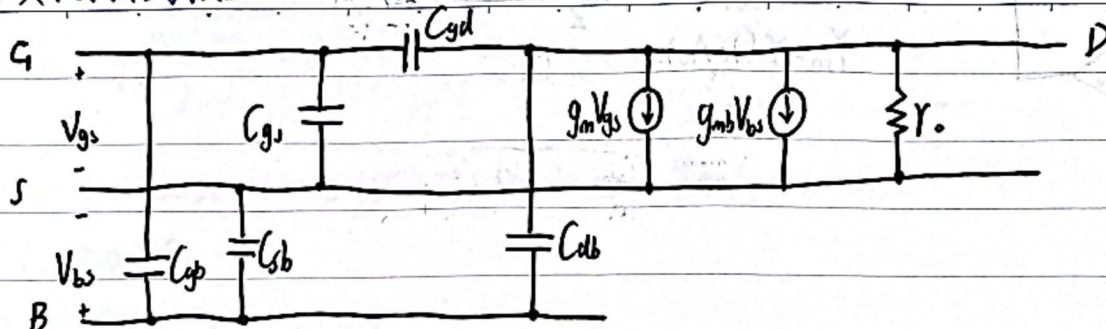
定义一个由于体效应导致的跨导

$$g_{mb} = \frac{\partial I_D}{\partial V_{BS}} = - \frac{\partial I_D}{\partial V_{SB}}$$

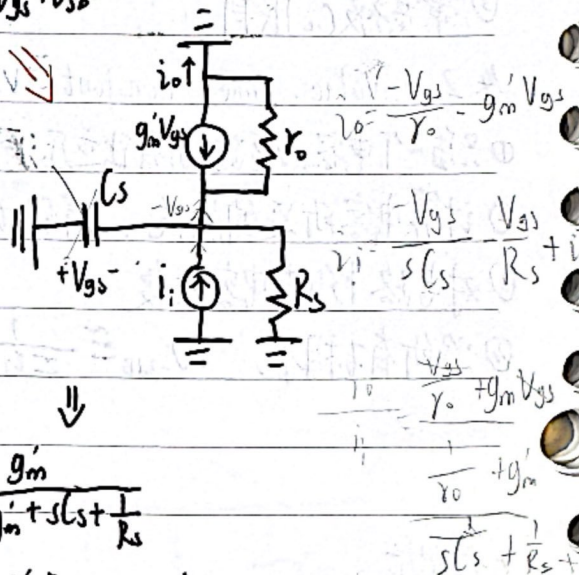
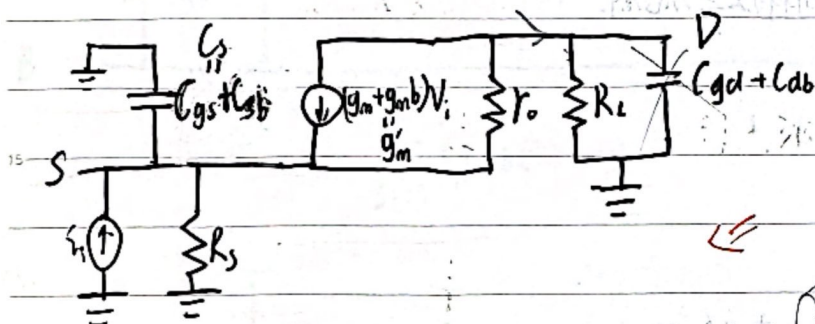
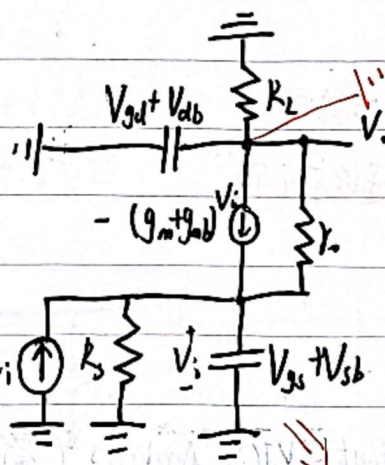
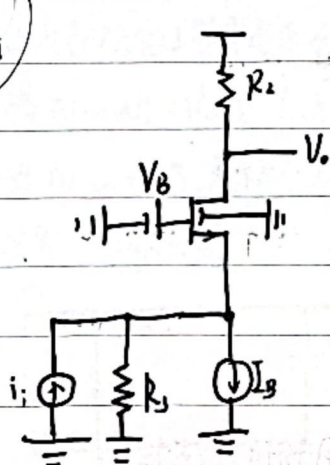
$$\frac{g_{mb}}{g_m} = - \frac{\partial V_t}{\partial V_{SB}} \cdot \frac{\partial I_D}{\partial V_t} \cdot \frac{\partial V_{GS}}{\partial I_D} = \frac{\partial V_t}{\partial V_{SB}} = \frac{\gamma}{2\sqrt{V_{SB} + 2\phi_f}}$$



2. 共栅小信号模型:



3.



$$\frac{i_o}{i_i} \approx \frac{g_m}{g_m + sC_s + \frac{1}{R_s}}$$

$$\approx \frac{g_m R_s}{1 + g_m R_s} \cdot \frac{1}{1 + s \frac{R_s C_s}{1 + g_m R_s}}$$

$$\text{当 } g_m R_s \gg 1 \text{ 时, } \frac{i_o}{i_i} \approx \frac{1}{1 + s \frac{C_s}{g_m}}$$

$$\text{低频时, } \frac{i_o}{i_i} \approx 1$$

输入阻抗

$$V_o \approx g_m (R_L \parallel r_o) V_{test}$$

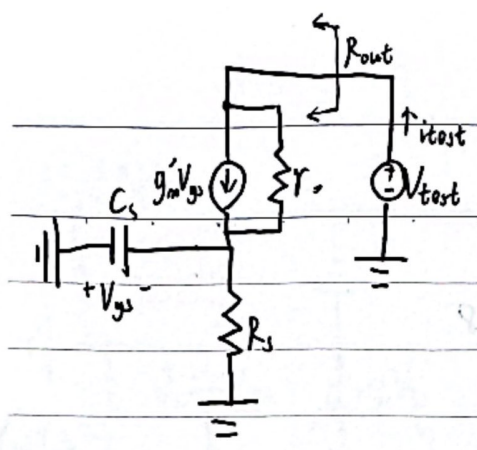
$$Y_{in} \approx \frac{g_m r_o}{R_L + r_o} + sC_s = \frac{g_m r_o}{R_L + r_o} (1 + sC_s \frac{R_L + r_o}{g_m r_o}) \quad \text{低频时, } R_{in} = \frac{1}{Y_{in}} \approx \frac{1}{g_m} (1 + \frac{R_L}{r_o})$$

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$$R_L \ll r_o \text{ 时 } R_{in} = \frac{1}{g_m}$$

$$R_L \gg r_o \text{ 时 } R_{in} = \frac{R_L}{g_m r_o}$$





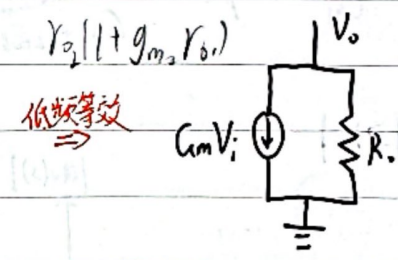
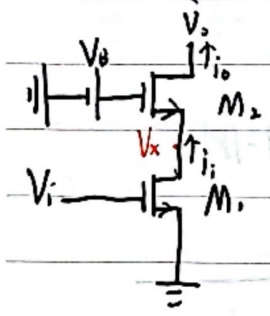
输出阻抗

$$i_{test} = \frac{V_{test}}{r_o} + \frac{V_{gs}}{R_s} + g_m' V_{gs}$$

$$V_{gs} = -i_{test} R_s$$

$$R_{out} = \frac{V_{test}}{i_{test}} \approx r_o (1 + g_m' R_s) \quad \text{当 } g_m' R_s \gg 1 \text{ 时, 非秋}$$

3. 共源共栅极



$$G_m = g_m \cdot \frac{i_o}{i_i} \approx g_m$$

$$R_o \approx r_{o2} (1 + g_m' r_{o1})$$

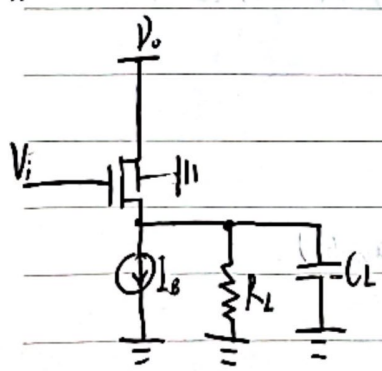
$$G_m R_o = g_m r_{o2} (1 + g_m' r_{o1}) \approx g_m r_{o1} \cdot g_m' r_{o2} \sim (g_m r_o)^2$$

$$\frac{V_o}{V_i} \approx g_m R_o \approx \frac{g_m}{g_m'} (1 + \frac{R_o}{r_{o1}}) \quad \text{接近 1}$$

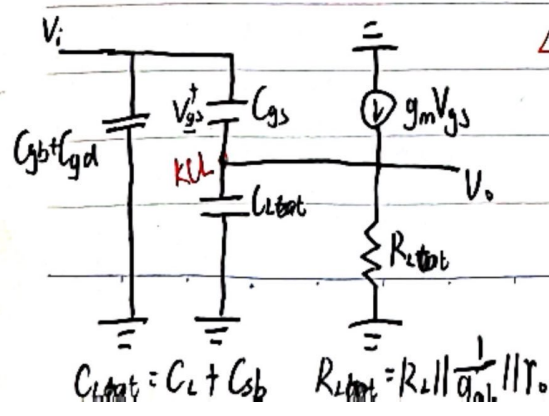
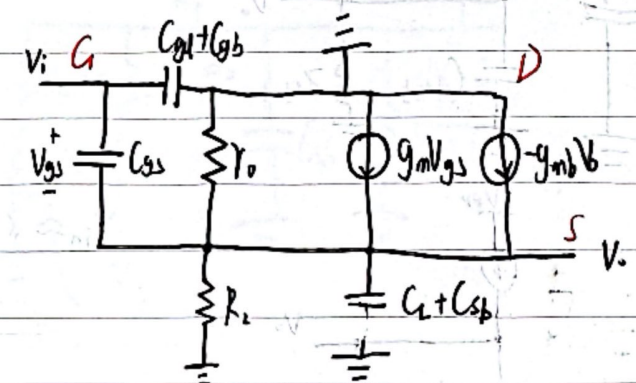
优点: 更接近 $g_m R_o$ 的电压增益, 更大带宽 (密勒效应减小)

缺点: 信号输出摆幅减小 (V_{gs2} 抬高, 电源电压不变, 动态范围减小)

共漏级放大器 (源极跟随器)



信号模型



$$V_o (s C_{tot} + s C_{gs} + \frac{1}{R_{Ltot}}) - V_i s C_{gs} - g_m (V_i - V_o) = 0$$

$$\frac{V_o}{V_i} = \frac{g_m + s C_{gs}}{g_m + s C_{gs} + s C_{Ltot} + \frac{1}{R_{Ltot}}}$$

电压传输函数

$$A_v = \frac{g_m}{g_m + \frac{1}{R_{Ltot}}} \cdot \frac{1 + \frac{s C_{gs}}{g_m}}{1 + \frac{s (C_{gs} + C_{Ltot})}{g_m + \frac{1}{R_{Ltot}}}}$$

$$A_{vo} = \frac{g_m}{g_m + \frac{1}{R_{tot}}}$$

$$R_{tot} = R_L \parallel \frac{1}{g_{mb}} \parallel r_o$$

• $R_L \rightarrow \infty, r_o \rightarrow \infty, g_{mb} = 0$

$$A_{vo} = 1$$

• $R_L \rightarrow \infty, r_o \rightarrow \infty, g_{mb} \neq 0$

$$A_{vo} = \frac{g_m}{g_m + g_{mb}} \approx 0.8$$

• $r_o \rightarrow \infty, g_{mb} = 0, R_L \text{ finite}$

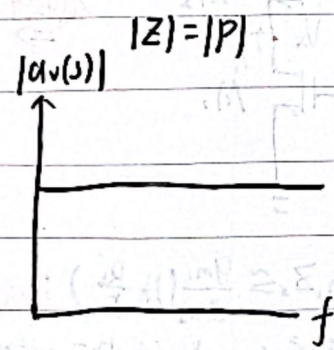
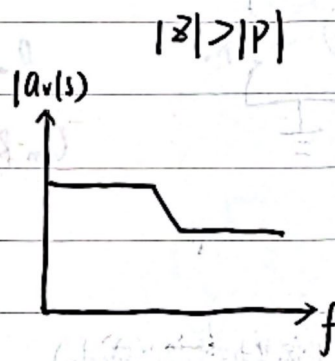
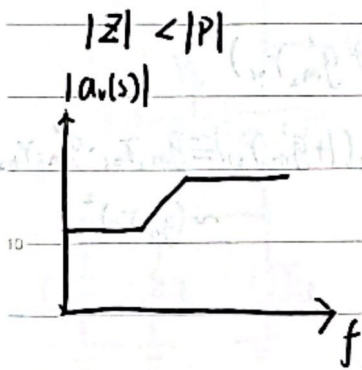
$$A_{vo} = \frac{g_m}{g_m + \frac{1}{R_L}}$$

高频下:

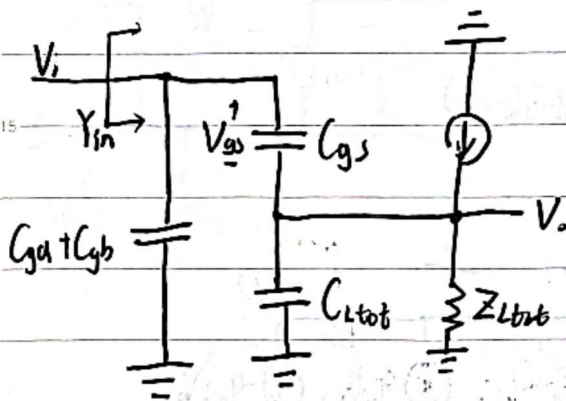
$$A_v(s) = \frac{V_o}{V_i} = A_{vo} \frac{1 - \frac{s}{Z}}{1 - \frac{s}{P}}$$

$$Z = -\frac{g_m}{C_{gs}}$$

$$P = -\frac{g_m + \frac{1}{R_{tot}}}{C_{gs} + C_{Ltot}}$$



输入阻抗

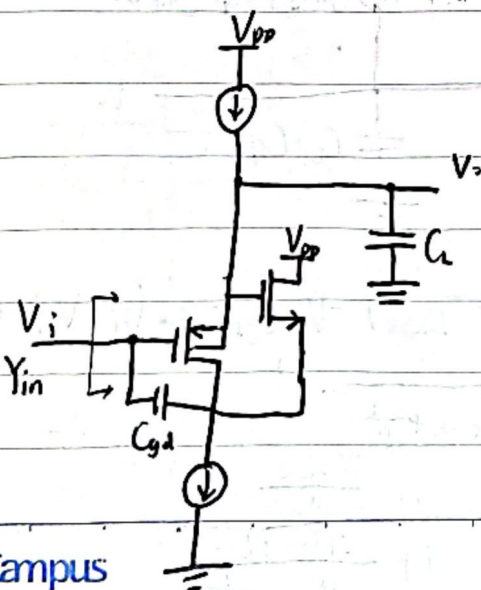


$$Y_{in} = s(C_{gd} + C_{gb}) + sC_{gs}(1 - A_v(s))$$

对于PMOS

$$Y_{in} = sC_{gd} + s(C_{gs} + C_{gb})(1 - A_v(s))$$

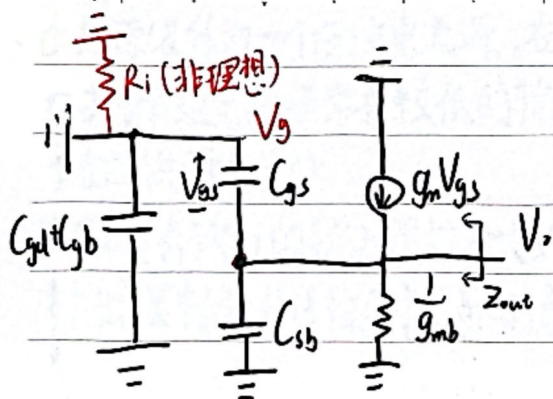
$$Y_{in} \approx sC_{gd}(1 - A_{vp}(s))A_{vn}(s)$$



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输出阻抗



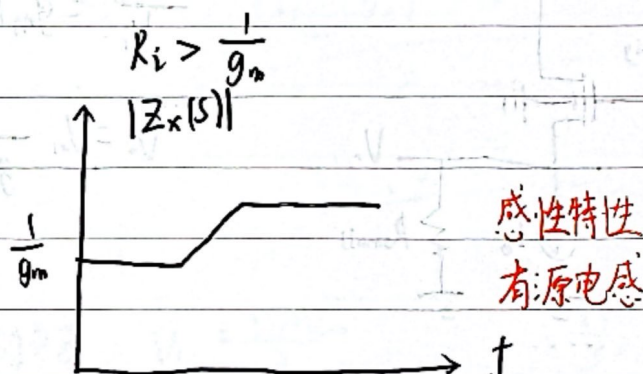
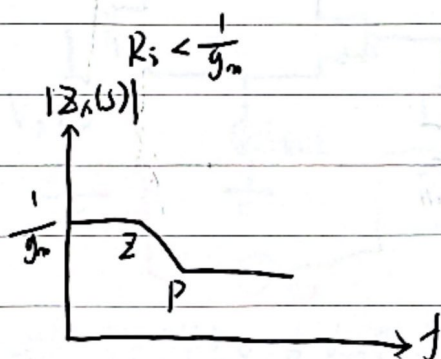
$$Z_{out} = \frac{1}{g_m + g_{mb}} \parallel \frac{1}{s(C_{gs} + C_{sb})}$$

$$Z_x = \frac{V_o}{i_x}$$

$$i_x = (V_o - V_g)(g_m + sC_{gs}) = V_o \left(1 - \frac{V_g}{V_o}\right)(g_m + sC_{gs})$$

$$V_g/V_o = \frac{R_i}{\frac{1}{sC_{gs}} + R_i}$$

$$Z_x \approx \frac{1}{g_m} \frac{1 + sR_iC_{gs}}{1 + \frac{sC_{gs}}{g_m}}$$

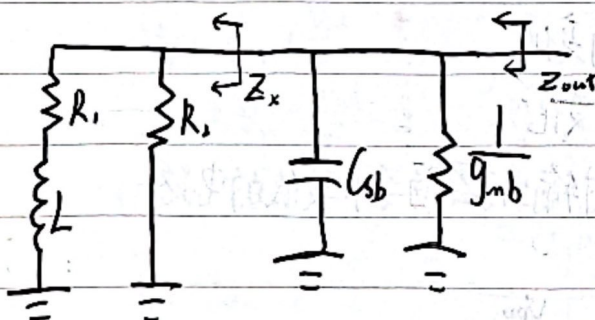


等效电路

$$R_1 \parallel R_2 = \frac{1}{g_m}$$

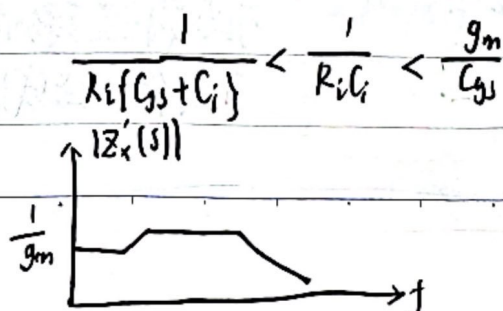
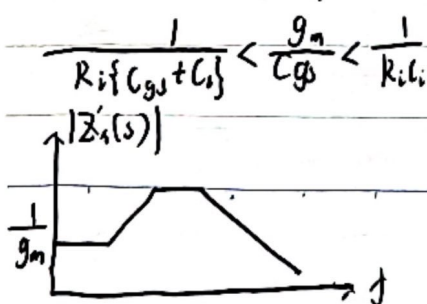
$$R_2 = R_i$$

$$L = \frac{R_i C_{gs}}{g_m R_i - 1}$$



考虑 $C_i = C_{gd} + C_{gb}$

$$Z'_x = Z_x \cdot \frac{C_{gs} + C_i}{1 + sR_iC_i}$$

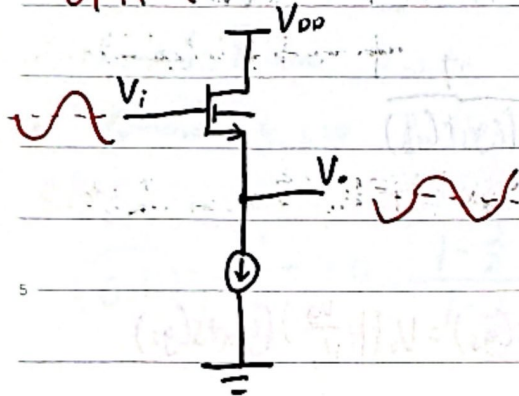


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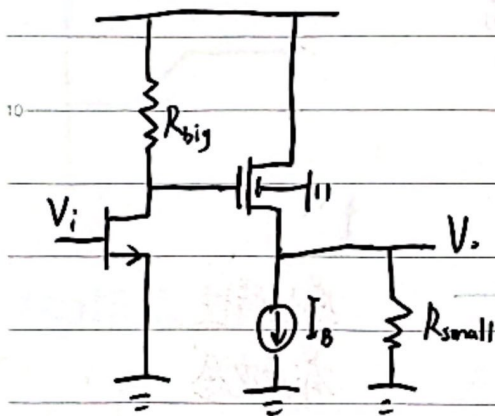


2. 共漏放大的应用:

① 电平转换

输出比输入低 $V_t + V_{ov}$

② 缓冲



$$\frac{V_{ov}}{V_i} = g_m R_{big}$$

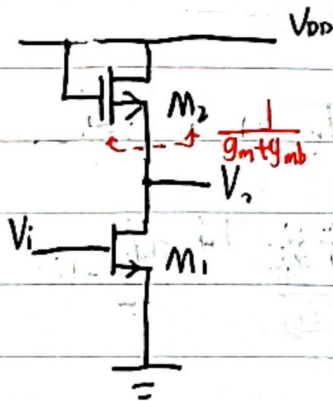
$$V_v = V_{ov} \frac{g_m}{g_m + g_{mb} + \frac{1}{R_{small}}}$$

3. 共漏的一些问题:

① V_t 随 V_v 变化而变化② I_D 和 V_{ov} 随 V_v 变化

③ 常常只能用于对输出摆幅要求较低的电路

④ 作为负载



$$\text{直流增益 } A_v = \frac{g_{m1}}{g_{m2} + g_{mb2}}$$

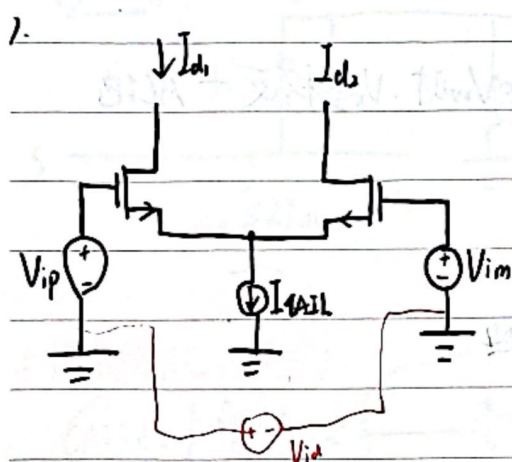
可以减小温度影响
可以补偿一些线性



4. 三种组态总结:

- ① 共源级作为一个压控电流源, 在输出接高阻抗负载时, 是一个很好的电压放大器
- ② 共栅级电路通常有较低的输入阻抗和较高的输出阻抗, 可以用于提高共源级的本征增益(共源共栅)
- ③ 共漏有较高的输入阻抗和较低的输出阻抗, 非常适合用于转换信号的直流工作点; 或在输出摆幅和非线性要求不高的情况下用作电压缓冲器

差分放大器

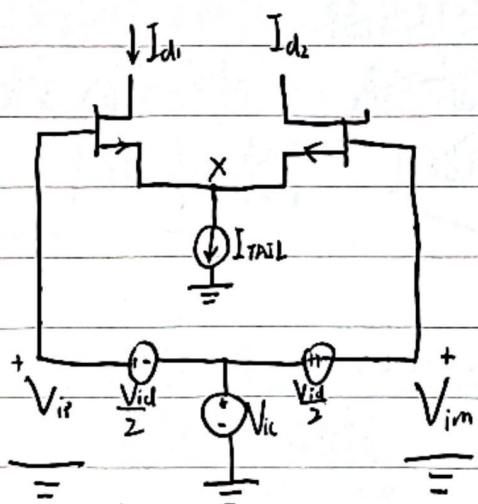


$$V_{id} = V_{ip} - V_{im}$$

$$\frac{(V_{gs1} - V_t)^2 \mu C_{ox} \frac{W}{L}}{2}$$

$$\frac{(V_{gs2} - V_t)^2 \mu C_{ox} \frac{W}{L}}{2}$$

共模: 描述节点 V_{ip} 和 V_{im} 相对于地 (GND) 的电压, $V_{ic} = \frac{V_{ip} + V_{im}}{2}$



$$V_{ip} = V_{ic} + \frac{V_{id}}{2}$$

$$V_{im} = V_{ic} - \frac{V_{id}}{2}$$

$$V_{ip} - V_{gs1} = V_{im} - V_{gs2}$$

$$I_{d1} + I_{d2} = I_{TAIL}$$

$$V_{gs1} = V_t + \sqrt{\frac{2I_{d1}}{\mu C_{ox} \frac{W}{L}}}$$

$$V_{gs2} = V_t + \sqrt{\frac{2I_{d2}}{\mu C_{ox} \frac{W}{L}}}$$

$$\frac{\mu C_{ox} \frac{W}{L}}{2} (V_{gs1} - V_{gs2}) (V_{gs1} + V_{gs2} - 2V_t)$$

$$\frac{\mu C_{ox} \frac{W}{L}}{2} V_{id}$$

$$V_{gs1} + V_{gs2} - 2V_t$$

$$(V_{gs1} - V_t) + (V_{gs2} - V_t)$$

$$V_{gs1} = V_t + \sqrt{\frac{2I_{d1}}{\mu C_{ox} \frac{W}{L}}}$$

$$V_{gs2} = V_t + \sqrt{\frac{2I_{d2}}{\mu C_{ox} \frac{W}{L}}}$$

$$I_{od} = I_{d1} - I_{d2} = \frac{1}{2} \mu C_{ox} \frac{W}{L} V_{id} \sqrt{\frac{4I_{TAIL}}{\mu C_{ox} \frac{W}{L}} - V_{id}^2}$$

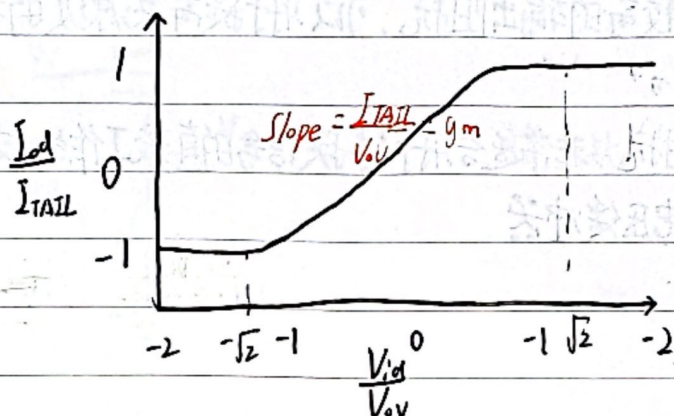
$$\therefore \frac{I_{TAIL}}{2} = \frac{1}{2} \mu C_{ox} \frac{W}{L} V_{ov}^2$$

$$\frac{I_{od}}{I_{TAIL}} = \frac{V_{id}}{V_{ov}} \sqrt{1 - \left(\frac{V_{id}}{2V_{ov}}\right)^2}$$



• 当 $V_{id}=0$ 时, $\frac{I_{od}}{I_{TAIL}}=0$

• 当 $V_{id}=\pm V_{OV}$ 时, 电流会流向一边 ($\frac{I_{od}}{I_{TAIL}}=\pm 1$)



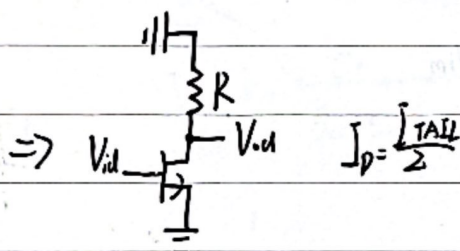
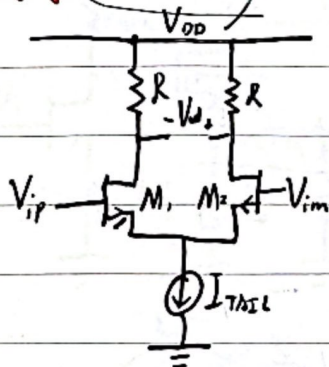
$$G_m = \left. \frac{dI_{od}}{dV_{id}} \right|_{V_{id}=0} = \frac{I_{TAIL}}{V_{OV}} = \frac{2I_D}{V_{OV}} = g_{m,2}$$

小信号跨导

对于x点, $V_x = V_{ic} - V_b - V_{OV} \sqrt{1 - \frac{1}{4} \left(\frac{V_{id}}{V_{OV}} \right)^2}$

当 $V_{id} \ll V_{OV}$ 时, V_x 基本不变 - AC地

差分对小信号等效电路



非线性特性

$$\frac{I_{od}}{I_{TAIL}} = \frac{V_{id}}{V_{OV}} \sqrt{1 - \left(\frac{V_{id}}{2V_{OV}} \right)^2} \xrightarrow{\text{泰勒展开}} \frac{I_{od}}{I_{TAIL}} \approx \frac{V_{id}}{V_{OV}} - \frac{1}{8} \left(\frac{V_{id}}{V_{OV}} \right)^3$$

当 $\frac{V_{id}}{V_{OV}} \ll 1$ 时, 线性度非常好 即 $V_{id} \ll V_{OV} = \frac{2}{\left(\frac{g_m}{I_{TAIL}} \right)}$

经验公式

对于1.5%的线性偏差, 需 $V_{id} < 0.5 \sqrt{\frac{2}{\left(\frac{g_m}{I_{TAIL}} \right)}}$

... 0.1% ... 0.2 ...

传输函数

差模增益 $A_{dm} = \frac{V_{od}}{V_{id}}$

共模增益 $A_{cm} = \frac{V_{oc}}{V_{ic}} \approx 0$



共模增益

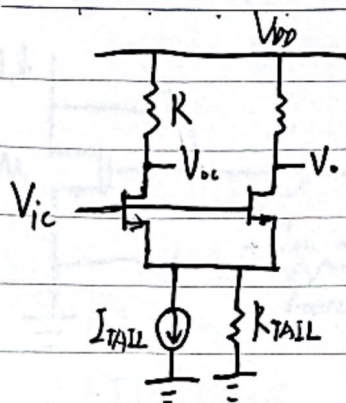
$$\frac{1}{2\sqrt{1+\beta}}$$

$$\beta V_{ov1} = \beta \Delta \beta V_{ov2}$$

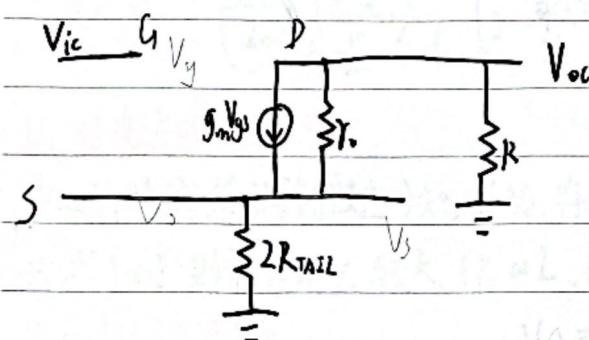
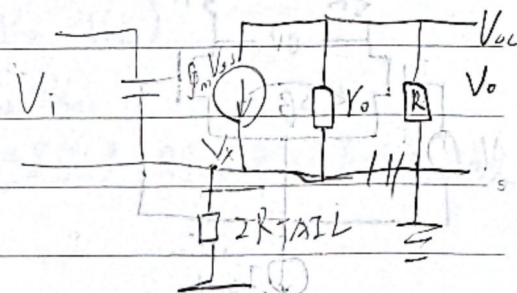
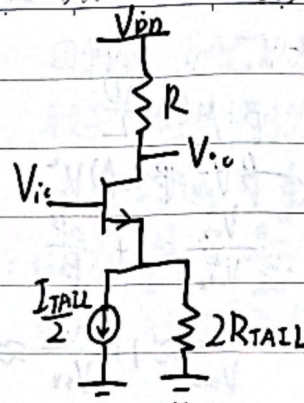
$$\frac{V_{ov1}}{V_{ov2}} = \frac{V_{ov1} V_{os}}{V_{ov2} V_{os}}$$

$$\frac{V_{ov1}^2}{V_{ov2}^2} = \frac{\beta \Delta \beta}{\beta}$$

$$\frac{V_{ov1}}{V_{ov2}} = \frac{V_{ov1}}{V_{ov2}} + \frac{V_{ov1}}{V_{ov2}} + g_m V_{ic}$$



=>



$$A_{cm} = \frac{V_{o1}}{V_{ic}} = - \frac{g_m}{1 + 2g_m R_{TAIL}} (R || r_o [1 + g_m 2R_{TAIL}])$$

$$\frac{V_{os}}{2R_{TAIL}} = \frac{V_{ic} - V_{os}}{V_{os}} g_m + \frac{V_{oc} - V_{os}}{V_{os}}$$

共模抑制比 (CMRR)

$$CMRR = \left| \frac{A_{dm}}{A_{cm}} \right| \approx \frac{g_m R}{1 + g_m 2R_{TAIL}} = 1 + g_m 2R_{TAIL}$$

$$\frac{V_{o1}}{R} + \frac{V_{o1} - V_{os}}{r_o} + g_m (V_{ic} - V_{os}) = 0$$

$$\frac{V_{oc}}{R} + \frac{V_{oc}}{r_o} + V_{ic} = V_{os} (g_m + \frac{1}{r_o})$$

电源抑制比 (PSRR): 输出对噪声的抑制能力

(由于半边电路不平衡导致)

$$A_v = \frac{V_{od}}{V_{dd}}$$

$$A_v \neq \frac{V_{od}}{V_{os}}$$

$$|PSRR_v| = \left| \frac{A_{dm}}{A_v} \right|$$

$$|PSRR_v| = \left| \frac{A_{dm}}{A_v} \right|$$

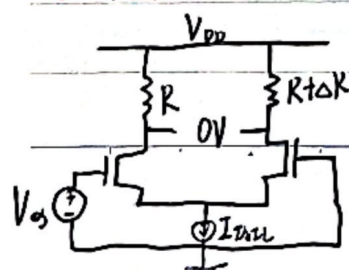
电流镜电路

1. 失调电压: 输入电压需为一个电压使得差分输出为0, 该电压称为失调电压

来源 ① 设计导致的系统误差

② 生产的随机误差

电阻不匹配



$$I_{D,R} = I_{D,R} (1 + \frac{\Delta R}{R})$$

$$1 + \frac{\Delta R}{R} = \frac{I_{D,R}}{I_{D,R}} = \frac{1 + \frac{1}{2} \frac{\Delta I_D}{I_D}}{1 - \frac{1}{2} \frac{\Delta I_D}{I_D}} \approx 1 + \frac{\Delta I_D}{I_D}$$

$$I_{D,R} = I_D + \frac{1}{2} \Delta I_D$$

$$I_{D,R} = I_D - \frac{1}{2} \Delta I_D$$

